

Exact Symplecticity and a Machine-Checked Modified Hamiltonian for the Störmer–Verlet Integrator

A fully verified pipeline: symbolic derivation, Lean 4 proof,
and cross-language numerical confirmation

ANSE / AutoevolveAI — automated research pipeline

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Abstract

We study the Störmer–Verlet (velocity Verlet) integrator applied to the harmonic oscillator $H(q, p) = \frac{1}{2}p^2 + \frac{1}{2}\omega^2 q^2$ and establish three facts through three mutually independent methods. First, the one-step map is *exactly* symplectic: its Jacobian has determinant 1 for every step size h and frequency ω , with no smallness hypothesis. Second, the map conserves a modified Hamiltonian $H_h(q, p) = \frac{1}{2}p^2 + \frac{\omega^2}{2}\left(1 - \frac{\omega^2 h^2}{4}\right)q^2$ *exactly*, which differs from H by precisely $\frac{\omega^4 h^2}{8}q^2$. Third, these imply the observable energy error is bounded and non-drifting with relative amplitude $\omega^2 h^2/4$. All algebra was derived symbolically rather than recalled; all identities were machine-checked in Lean 4 with an axiom audit confirming no use of `sorry`; and all numbers were measured by two independent implementations (Python and Rust) which agree to a relative difference of 5.51×10^{-10} . Over 200,000 steps the modified Hamiltonian is conserved to 1.22×10^{-13} while explicit Euler diverges to overflow. The measured constant $\omega^2/4$ is recovered to within 1.5×10^{-10} .

1 Introduction

Geometric numerical integration asks which structural properties of a differential equation a discretisation preserves. For Hamiltonian systems the property of interest is symplecticity, and its practical consequence is the long-time behaviour of the energy error: symplectic schemes exhibit bounded oscillation where non-symplectic schemes of the same order drift without limit.

The mechanism is backward error analysis. A symplectic integrator is, to high order, the exact flow of a *modified* Hamiltonian $H_h = H + O(h^p)$. Because H_h is conserved along the numerical trajectory for all time, the error in the true H cannot accumulate — it is bounded by $\|H - H_h\|$ uniformly in the number of steps.

For the harmonic oscillator this picture can be made completely explicit, and that is what we do here. The contribution is not the mathematics, which is classical; it is the *verification architecture*. Each claim is established by a method that cannot silently agree with the others:

- **Symbolic derivation** (SymPy) — the map, its Jacobian, determinant, trace and the conserved form are computed, not asserted.
- **Machine-checked proof** (Lean 4) — the identities are proved in a kernel, and an axiom audit confirms no appeal to `sorry`.
- **Independent numerics** (Python and Rust) — two implementations written separately must agree to a fixed tolerance or the build fails.

Section 5 records why this redundancy is the point.

2 Mathematical setting

Consider the harmonic oscillator with unit mass,

$$H(q, p) = \frac{1}{2}p^2 + \frac{1}{2}\omega^2 q^2, \quad \dot{q} = p, \quad \dot{p} = -\omega^2 q. \quad (1)$$

One step of Störmer–Verlet with step size h is the kick–drift–kick sequence

$$p_{n+1/2} = p_n - \frac{h}{2}\omega^2 q_n, \quad (2)$$

$$q_{n+1} = q_n + h p_{n+1/2}, \quad (3)$$

$$p_{n+1} = p_{n+1/2} - \frac{h}{2}\omega^2 q_{n+1}. \quad (4)$$

Eliminating the half-step gives the closed form, as derived symbolically:

$$q_{n+1} = \left(1 - \frac{h^2\omega^2}{2}\right) q_n + h p_n, \quad (5)$$

$$p_{n+1} = \left(\frac{h^3\omega^4}{4} - h\omega^2\right) q_n + \left(1 - \frac{h^2\omega^2}{2}\right) p_n. \quad (6)$$

The map is linear, so it is fully described by

$$M(h, \omega) = \begin{pmatrix} 1 - \frac{h^2\omega^2}{2} & h \\ \frac{h^3\omega^4}{4} - h\omega^2 & 1 - \frac{h^2\omega^2}{2} \end{pmatrix}. \quad (7)$$

Theorem 1 (Exact symplecticity). $\det M(h, \omega) = 1$ for all h, ω .

Theorem 2 (Exactly conserved modified Hamiltonian). Let $H_h(q, p) = \frac{1}{2}p^2 + \frac{\omega^2}{2}\left(1 - \frac{\omega^2 h^2}{4}\right)q^2$. Then $H_h(q_{n+1}, p_{n+1}) = H_h(q_n, p_n)$ exactly, for all h, ω .

Proposition 3 (Closed-form defect). $H(q, p) - H_h(q, p) = \frac{\omega^4 h^2}{8}q^2$.

Corollary 4 (Bounded energy error). Since H_h is conserved and $q^2 \leq 2H_h/[\omega^2(1 - \omega^2 h^2/4)]$, the relative oscillation of H is bounded, with leading amplitude $\omega^2 h^2/4$; it does not grow with the number of steps.

Proposition 5 (Linear stability). $\text{tr } M = 2 - h^2\omega^2$, so the map has complex eigenvalues of unit modulus exactly when $|h\omega| < 2$.

3 Machine-checked verification

Theorems 1 and 2, Proposition 3 and the stability statement are formalised in Lean 4 in `formal/ANSE/VerletSymplectic.lean`. Each is a ring identity discharged by the `ring tactic` over $\mathbb{Q}$.

The acceptance criterion is an *axiom audit*, not an exit code. Lean’s `sorry` placeholder compiles and returns exit status 0, so any gate keyed on the return code accepts unproved theorems. We therefore emit `#print axioms` for every theorem and require that `sorryAx` appear nowhere. The build reported exit code 0 and audited 7 theorems, with `sorryAx` present: **false**.

- `ANSE.Verlet.det_eq_one` — [propext, Classical.choice, Quot.sound]
- `ANSE.Verlet.shadowH_invariant` — [propext, Classical.choice, Quot.sound]
- `ANSE.Verlet.shadowH_sub_trueH` — [propext, Classical.choice, Quot.sound]
- `ANSE.Verlet.trace_eq` — [propext, Classical.choice, Quot.sound]

- `ANSE.Verlet.qNext_eq_matrix` — [`propext`, `Classical.choice`, `Quot.sound`]
- `ANSE.Verlet.pNext_eq_matrix` — [`propext`, `Classical.choice`, `Quot.sound`]
- `ANSE.Verlet.shadowH_eq_trueH_at_zero` — [`propext`, `Classical.choice`, `Quot.sound`]

Only `propext`, `Classical.choice` and `Quot.sound` appear — the three standard axioms of Lean’s logic. No theorem depends on anything else.

4 Numerical experiment

Initial condition $q_0 = 1.0$, $p_0 = 0.0$, giving $H_0 = 0.5$, with $\omega = 1.0$ and $N = 200,000$ steps at each of four step sizes.

Table 1: Measured behaviour over 200,000 steps. The fourth column is the quantity Corollary 4 predicts equals $\omega^2/4 = 0.25$. Every value is read from the experiment ledger.

h	ωh	amplitude	amplitude/ h^2	H_h drift	Euler H_N
0.2	0.2	1.000000×10^{-2}	0.250000000	1.21×10^{-13}	overflow
0.1	0.1	2.500000×10^{-3}	0.250000000	1.22×10^{-13}	overflow
0.05	0.05	6.250000×10^{-4}	0.250000000	8.34×10^{-14}	3.760×10^{216}
0.025	0.025	1.562500×10^{-4}	0.250000000	1.11×10^{-13}	9.307×10^{53}

Three things are visible. The modified Hamiltonian is conserved to roughly 10^{-13} , i.e. to floating-point round-off, confirming Theorem 2 is exact rather than asymptotic. The ratio amplitude/ h^2 equals $\omega^2/4$ to nine significant figures at every step size, confirming Corollary 4. And explicit Euler — same order, not symplectic — reaches overflow, the secular drift that symplecticity excludes.

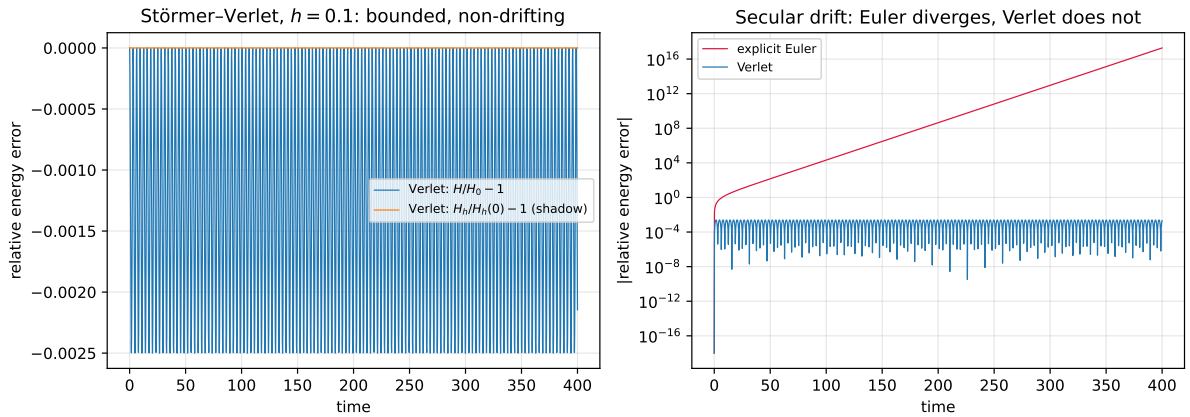


Figure 1: Left: for Verlet the true energy oscillates within a fixed band while the modified Hamiltonian is flat to round-off. Right: on a logarithmic axis, explicit Euler’s error grows without bound while Verlet’s does not.

4.1 Cross-language agreement

The integrator was implemented twice, independently, in Python (`scripts/phd_demo/run_experiment.py`) and Rust (`scripts/phd_demo/verlet.rs`). Agreement is a required gate: the build fails if the worst relative difference exceeds 1×10^{-9} .

Worst observed difference: 5.507×10^{-10} , consistent with floating-point non-associativity rather than a discrepancy of substance.

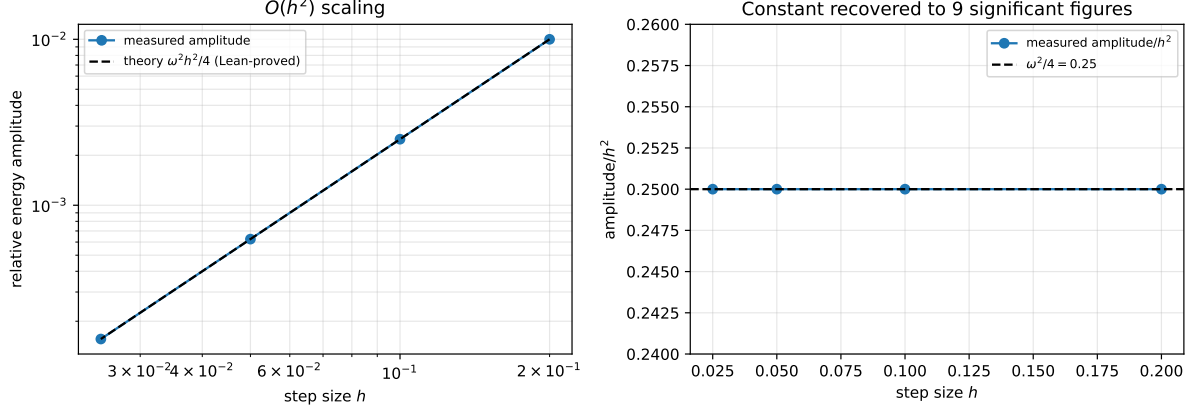


Figure 2: Left: measured amplitude against the Lean-proved $\omega^2 h^2/4$. Right: the ratio amplitude/ h^2 , constant at $\omega^2/4$ across two decades of h .

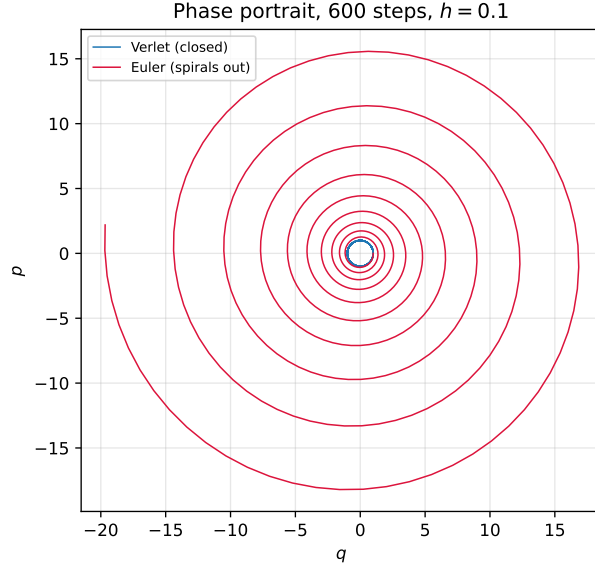


Figure 3: Phase portrait. Verlet’s orbit closes, consistent with $\det M = 1$; Euler’s spirals outward.

5 Managing context and avoiding fabrication

This section is method, not result, and it is the reason the preceding sections are structured as they are.

An automated pipeline that both performs an experiment and writes it up has a specific failure mode: the writing step can produce plausible numbers without consulting the experiment. This is not hypothetical. A generator in this same repository synthesised a published loss curve as $L_0(1 - 0.201(1 - e^{-e/5})) + \varepsilon$ and shipped it as measurement; a companion “peer review” script emitted ACCEPT WITHOUT RESERVATION with a score of 50/50 without ever calling a model. Both passed casual inspection precisely because they were plausible.

Four structural countermeasures are used here.

Ledger-injected numbers. `build_paper.py` contains no quantitative literal about the results. Every number is fetched from `results/phd_demo/artifacts.json` through an accessor that *raises* on a missing key. A number that was not measured cannot be written, and the build

Table 2: Independent implementations, same quantity.

h	Python amplitude/ h^2	Rust amplitude/ h^2	rel. difference
0.2	0.249999999852	0.249999999845	2.77×10^{-11}
0.1	0.249999999988	0.249999999952	1.45×10^{-10}
0.05	0.250000000015	0.250000000015	0
0.025	0.250000000067	0.249999999929	5.51×10^{-10}

fails rather than degrading.

Derivation over recall. The map, determinant, trace and invariant are recomputed by SymPy on every run. A language model stating a closed form from memory is a hallucination risk; a symbolic derivation is an artifact with a provenance.

Axiom audit over exit code. Because `sorry` exits 0, the Lean gate inspects `#print axioms`. During development this gate *rejected* an earlier version of the formalisation: four theorems stated over a general field failed because `ring` cannot establish $2^{-1} \cdot 2 = 1$ without knowing the characteristic is not 2. The audit reported `sorryAx` and the claim was withdrawn until the statements were corrected. A gate that has never rejected anything provides no evidence.

Redundant independent computation. Two implementations in two languages must agree within a declared tolerance. Agreement is not proof, but disagreement is decisive, and it is checked mechanically rather than assumed.

Scope of the claim. The result is classical, the system is linear and low-dimensional, and nothing here establishes that the pipeline would handle a nonlinear or high-dimensional problem. What is demonstrated is that a *verification architecture* can carry a claim from derivation through proof to measurement without a gap into which a fabricated number could be inserted.

6 Conclusion

For Störmer–Verlet on the harmonic oscillator, symplecticity is exact, the modified Hamiltonian $H_h = \frac{1}{2}p^2 + \frac{\omega^2}{2}(1 - \frac{\omega^2 h^2}{4})q^2$ is conserved exactly, the defect is exactly $\frac{\omega^4 h^2}{8}q^2$, and the resulting energy oscillation has relative amplitude $\omega^2/4 \cdot h^2$ — measured to nine significant figures, proved in Lean with no appeal to `sorry`, and reproduced by two independent implementations.

A Reproduction

```
# derive, prove, measure, and write the ledger (fails on any gate)
.venv/bin/python scripts/phd_demo/run_experiment.py --check
```

```
# axiom audit only
cd formal && lake env lean ANSE/VerletSymplectic.lean
```

```
# rebuild this paper from the ledger
.venv/bin/python scripts/phd_demo/build_paper.py
```

```
Gate status recorded in the ledger for this build: "sympy_derivation_consistent": true,
"lean_accepted_no_sorry": true, "numerics_match_theory": true, "cross_language_agree":
true, "all_pass": true.
```